

MATSim User Survey

MUM 2026 · Paris

60 Responses

Only complete surveys were collected.

0

43

10

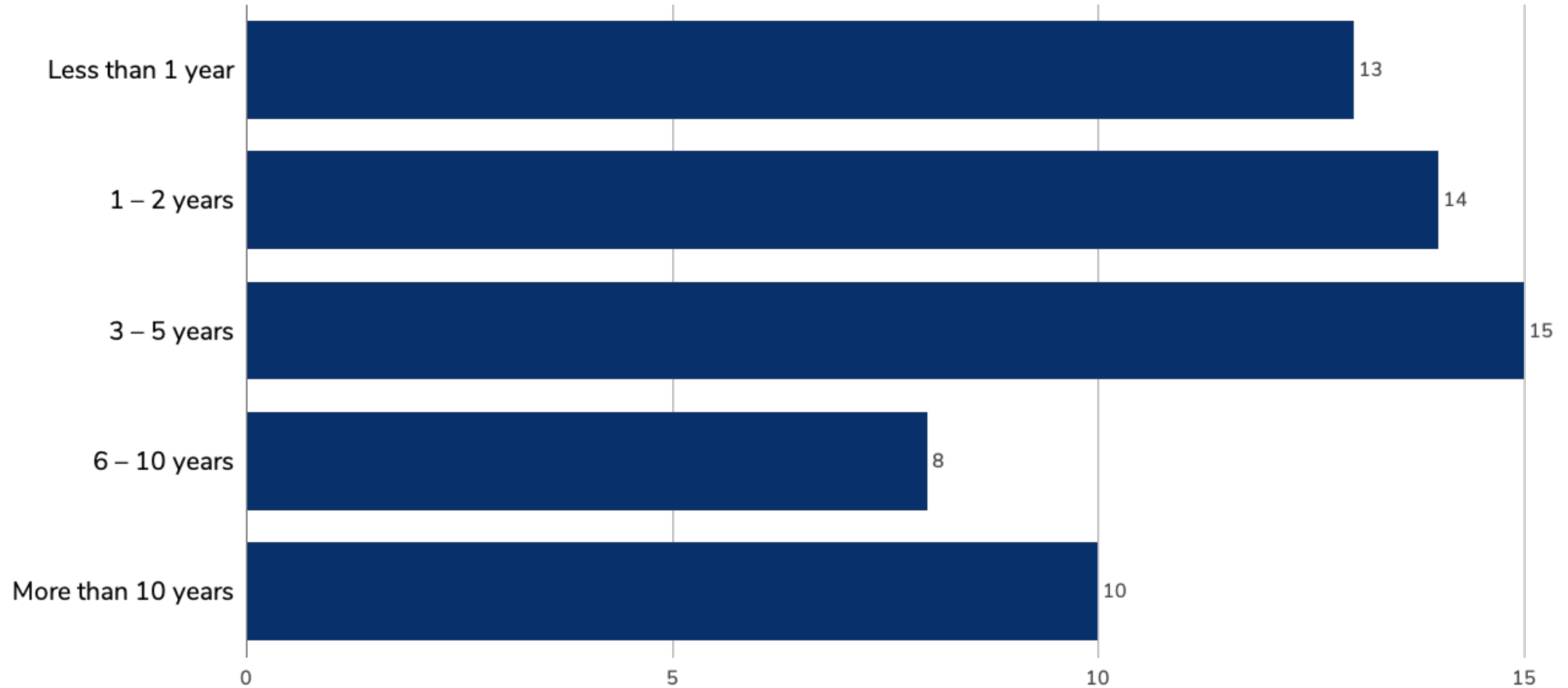
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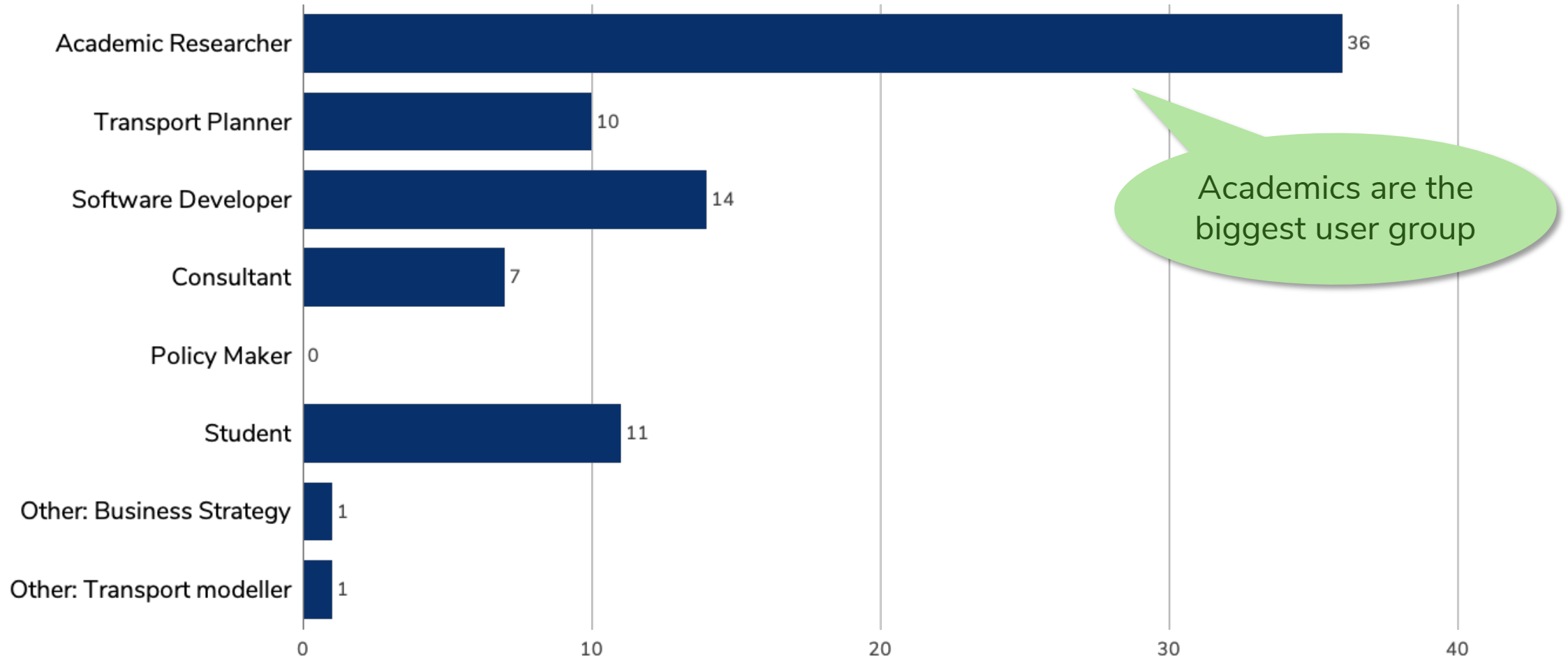
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Nobody from
America?

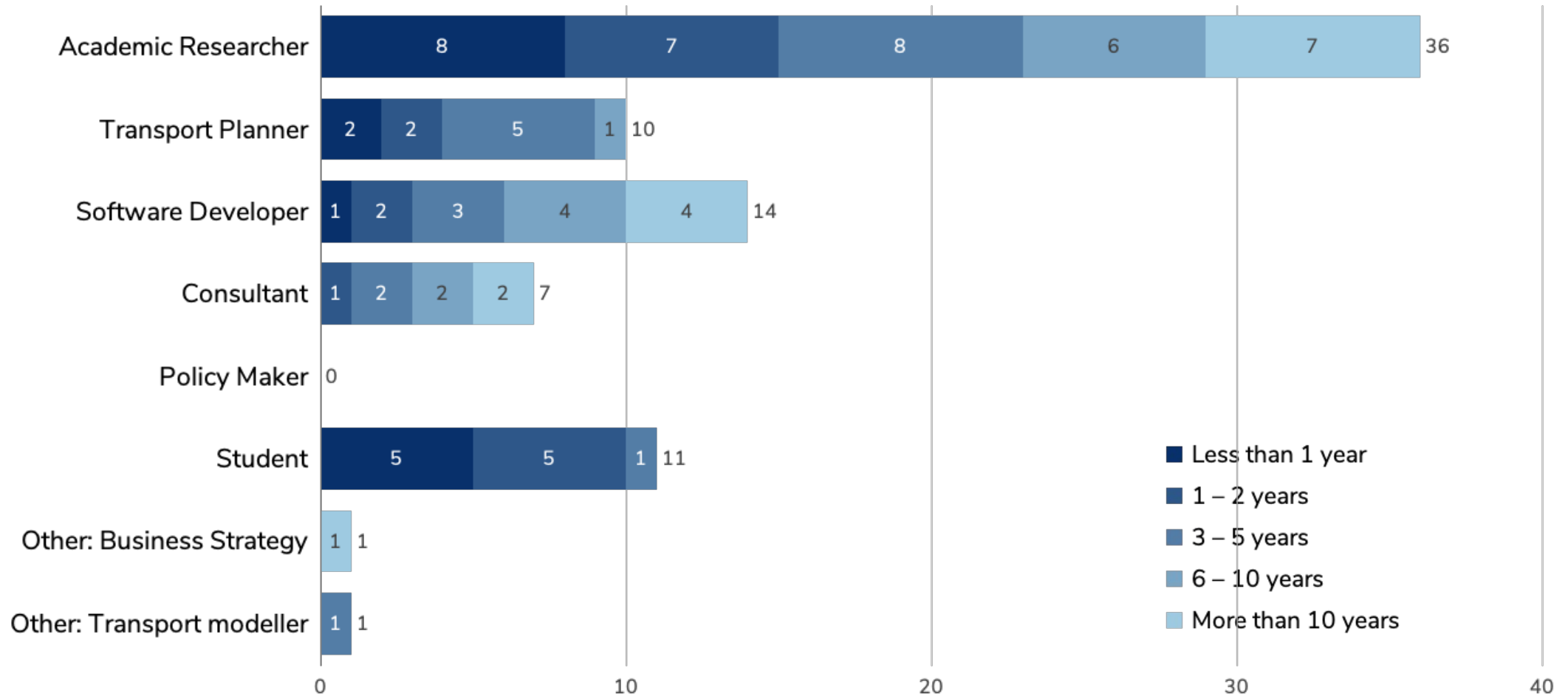
How long have you been using MATSim?



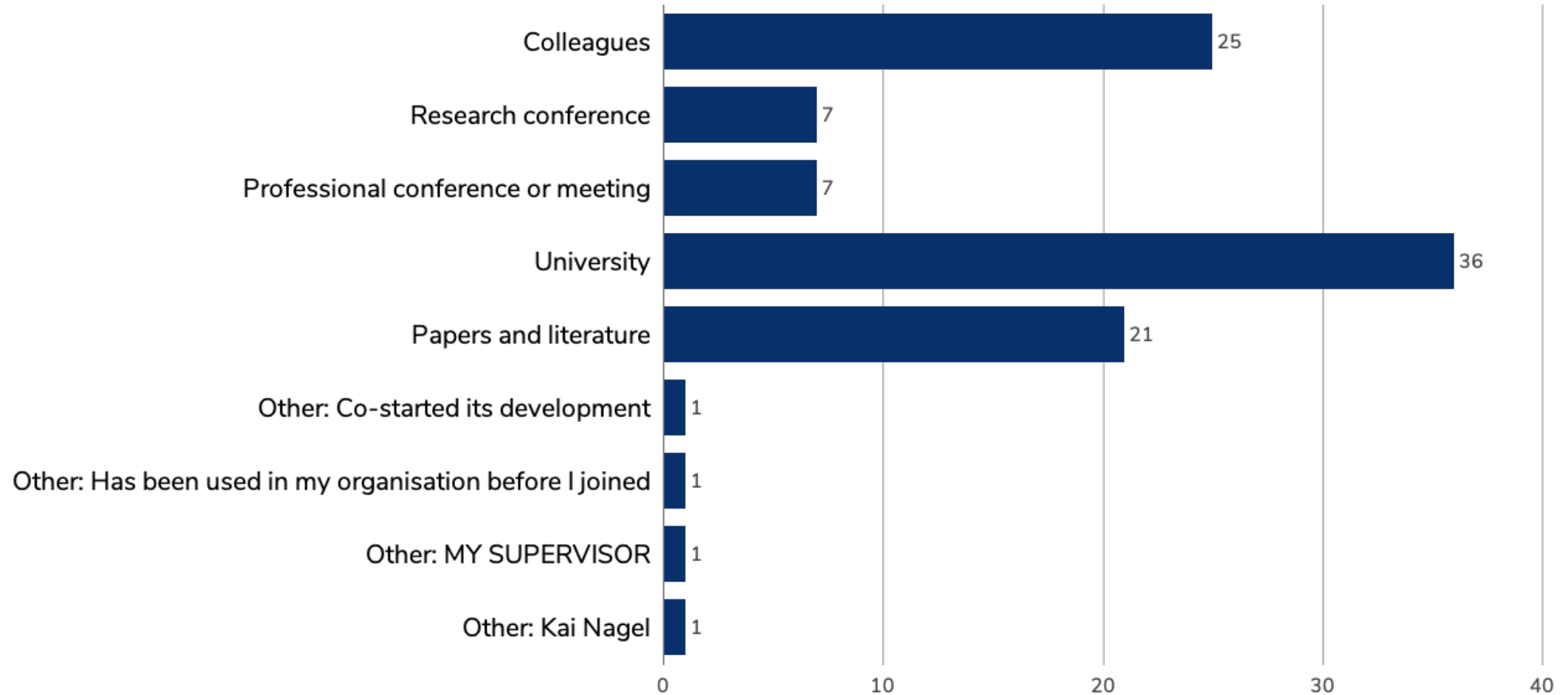
What is your primary role in relation to MATSim?



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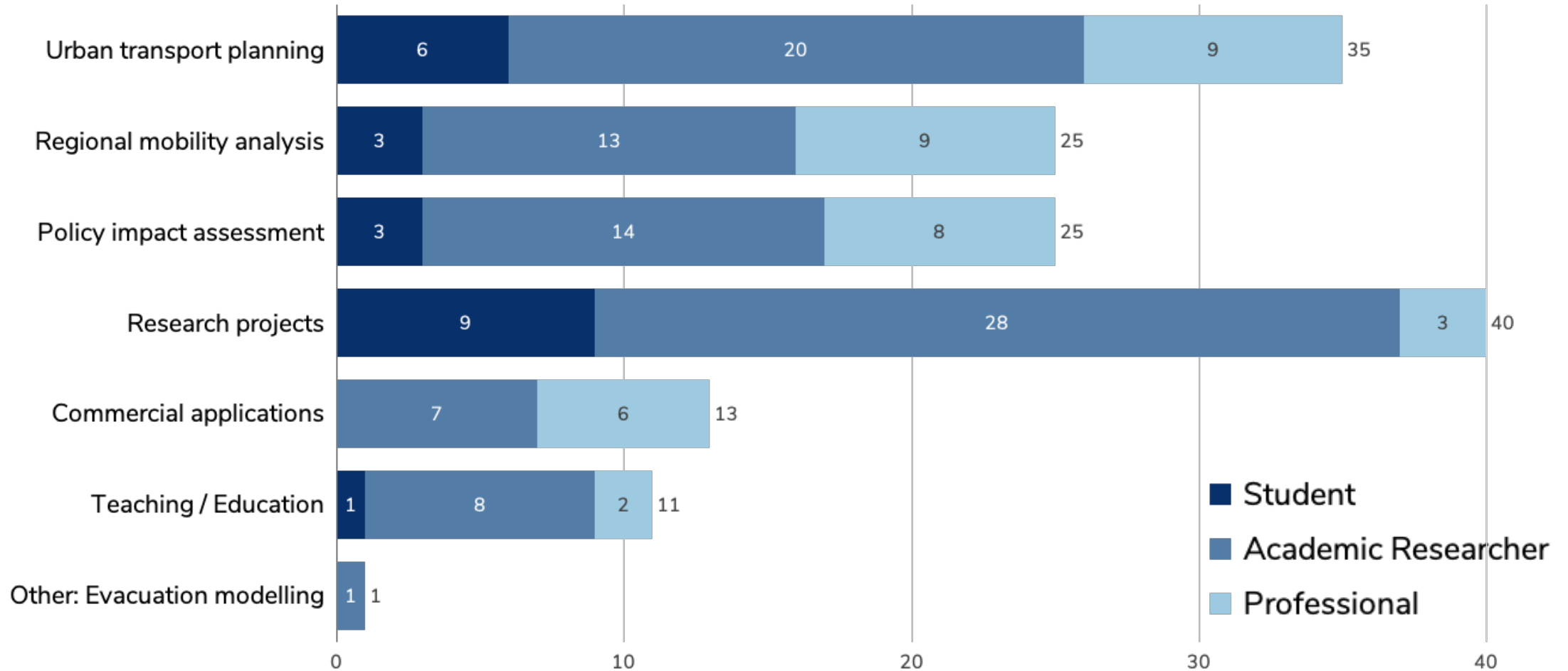


Where did you hear/learn about MATSim?

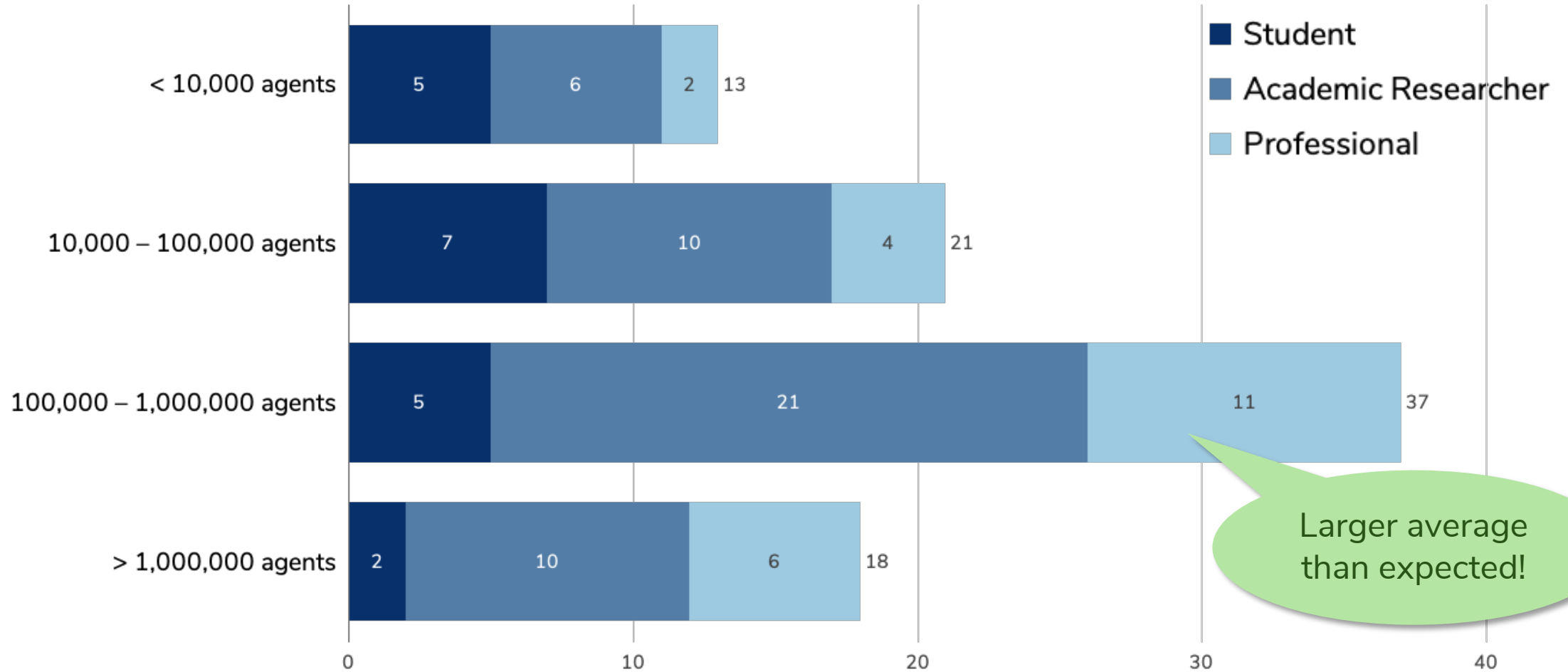


Usage Patterns

What type of projects do you use MATSim for?

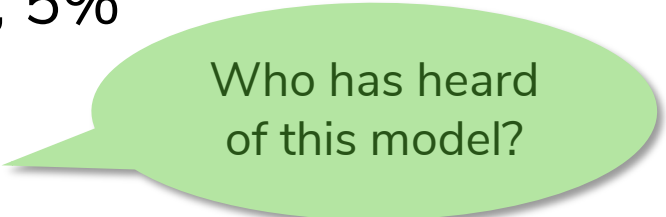


What is your typical simulation size?



Describe your most important simulations

- Berlin, 500k agents (10%)
- Switzerland, 8m agents
- Singapore, 7m agents
- Île-de-France, 12m agents
- Greater Melbourne
- Corsica, ~350k agents
- Hampton Roads region in Virginia, USA, 1.8m pop size, 5%
- Large-scale Evacuation Models (Australia)
- Whole of New Zealand main islands, 1m agents, 25%
- Brno Metropolitan Region, 900k agents, 100%
- ... and more!

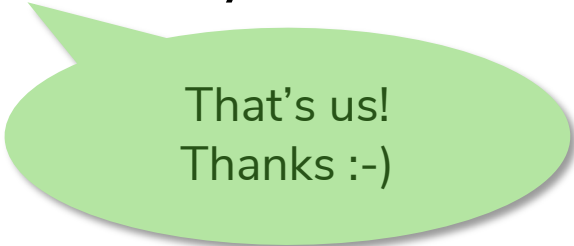


Who has heard
of this model?

What made you choose MATSim initially?

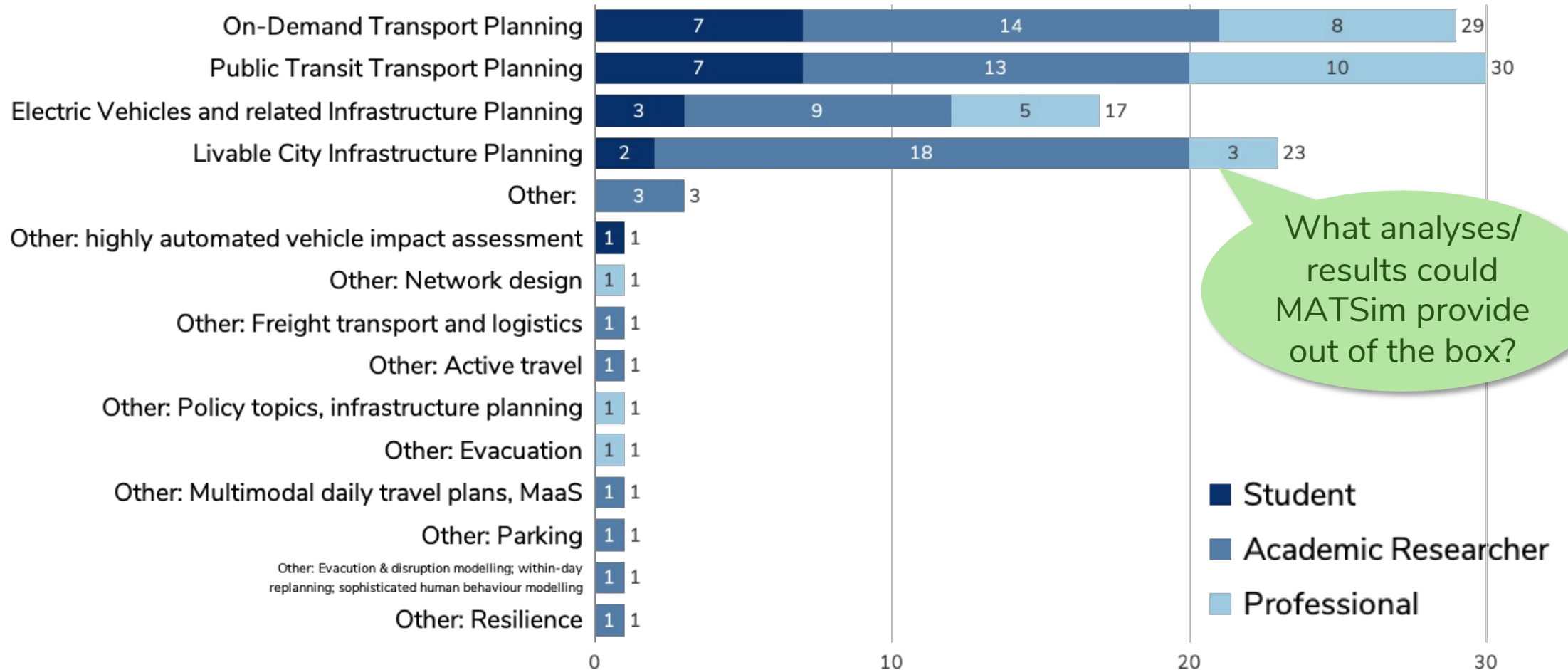
Why not any other framework or tool?

- Open Source
- MATSim allows more flexibility
- I fancy the framework of MATSim, it is one of the most straight forward way to model MaaS and impact of policy changes.
- It is a very powerful tool.
- Is tough to learn but really rewarding in the things that it can do.
- Suitability for large scale simulations, ability to handle large simulations relatively fast
- Being researchers developed and maintained, great community
- Overwhelming literature accessibility and papers
- Replanning capabilities, modularity



That's us!
Thanks :-)

Which topics will become more important in your work in the next few years?

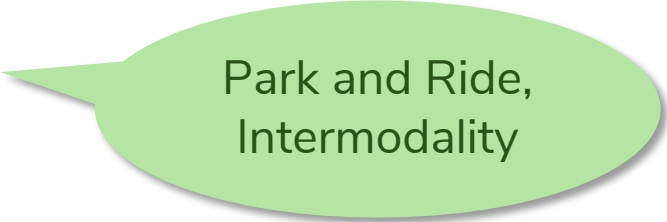


What new features or enhancements would most likely convince you to invest further in MATSim or upgrade your current use? (1/4)

- Everything must be fast, fast, fast!
 - Improvements in performance
 - Accelerate simulation
 - Better memory managements
-
- better represent multi-modal trips
 - Link to time-use models
 - Better parking models
 - Modules such park-and-ride



Performance



Park and Ride,
Intermodality

What new features or enhancements would most likely convince you to invest further in MATSim or upgrade your current use? (2/4)

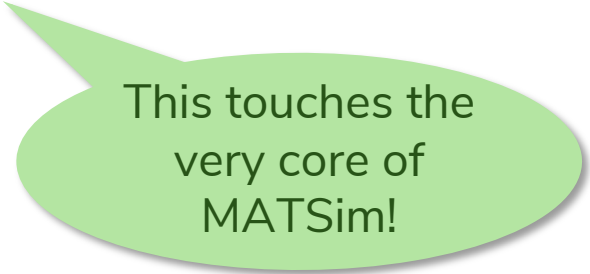


Documentation
Documentation
Documentation

- More documentation
- Better documentation
- Slightly better/more user-friendly interface and documentation
- Better visibility on new developments and functionalities (application contrib, DSim)
- Clearer and modular structure (interoperability of all components)
- UI for easy and foolproof pom.xml and config generation with default values based on the contribs one includes in a simulation.
- Stronger community support would motivate me to invest further in MATSim especially better forums, clearer documentation, and more practical examples to make troubleshooting and learning easier.

What new features or enhancements would most likely convince you to invest further in MATSim or upgrade your current use? (3/4)

- Move on from 90s selling points (everybody does agents by now), adopt more state of the art mainstream concepts, eg in terms of traffic assignment, choice set generation, statistical techniques.
- Better traffic model (other options than the Queue model). Intersection model.



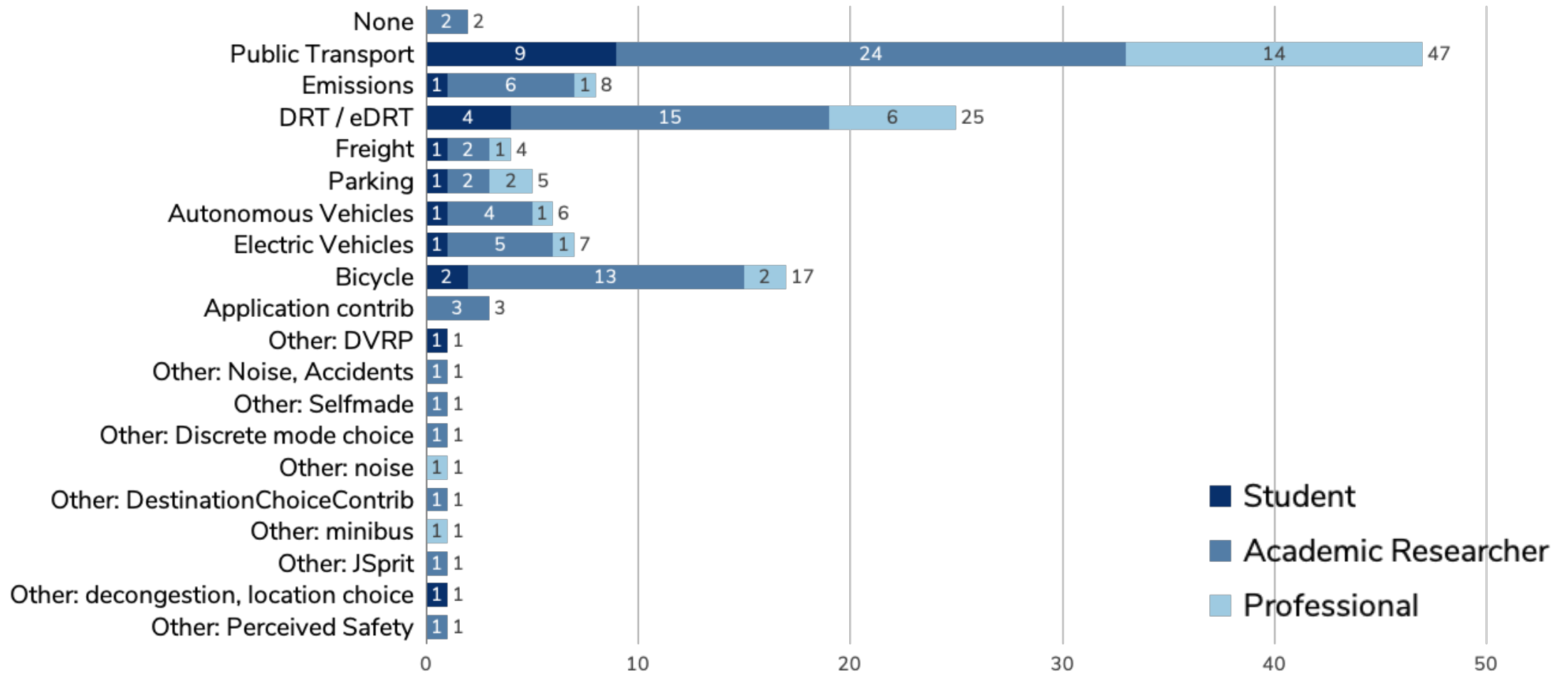
This touches the
very core of
MATSim!

What new features or enhancements would most likely convince you to invest further in MATSim or upgrade your current use? (4/4)

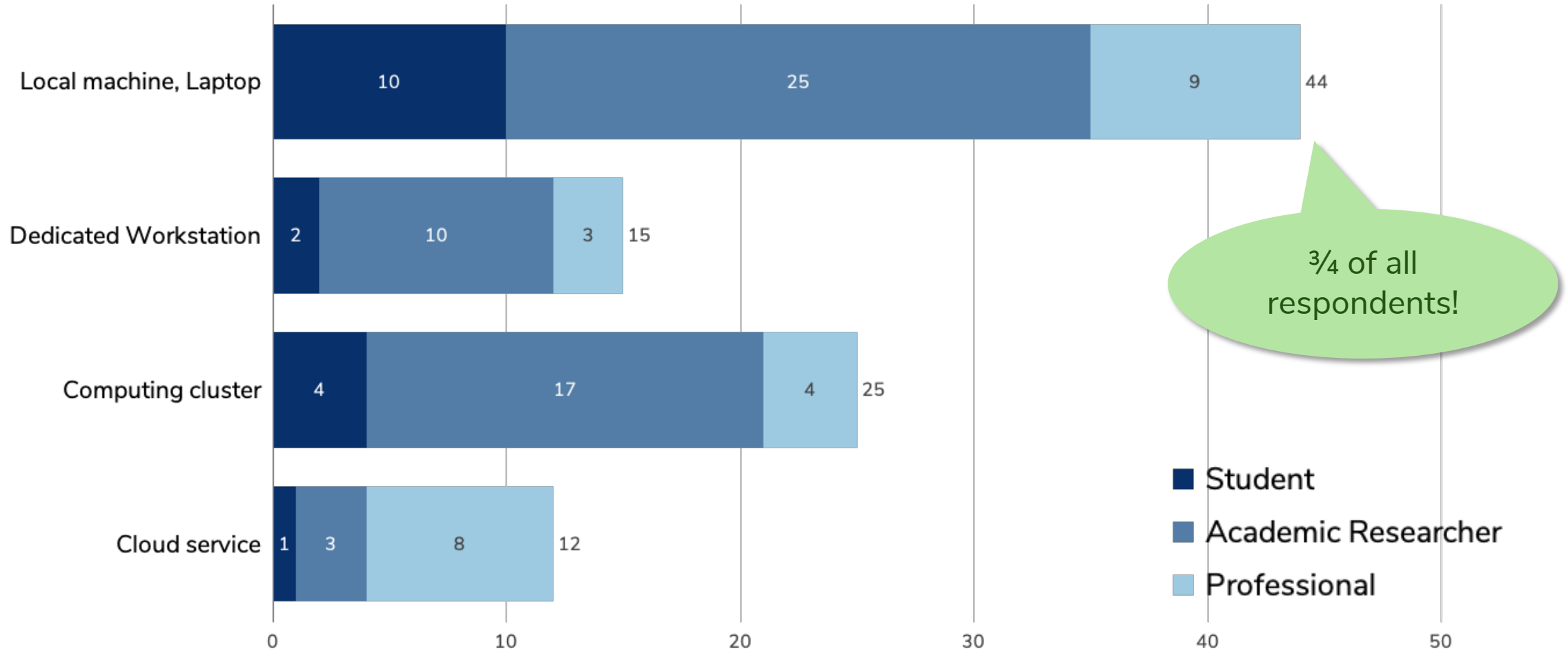
- Moving away from Java
- Fully compatible Python API.
- Active community
- Linkage to AI

Technical Details

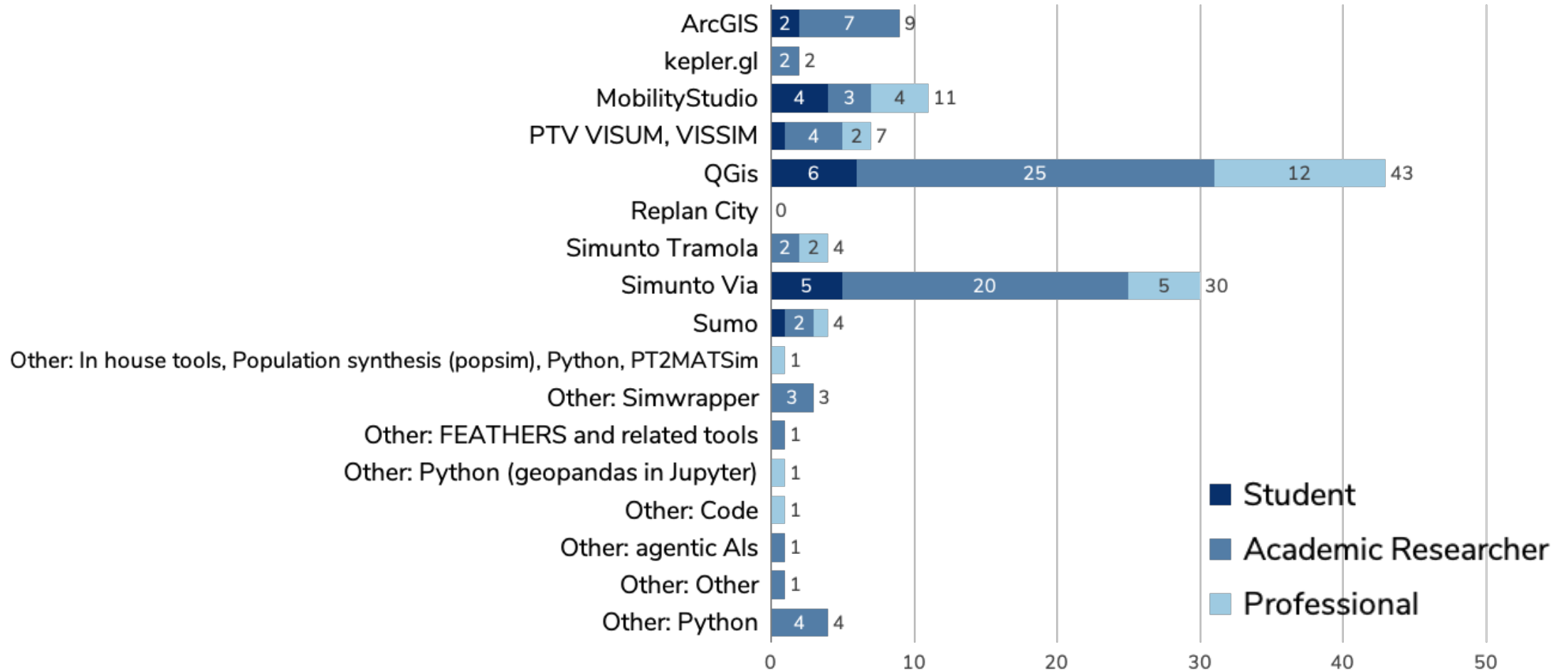
Which MATSim extensions do you regularly use?



How do you typically run MATSim?

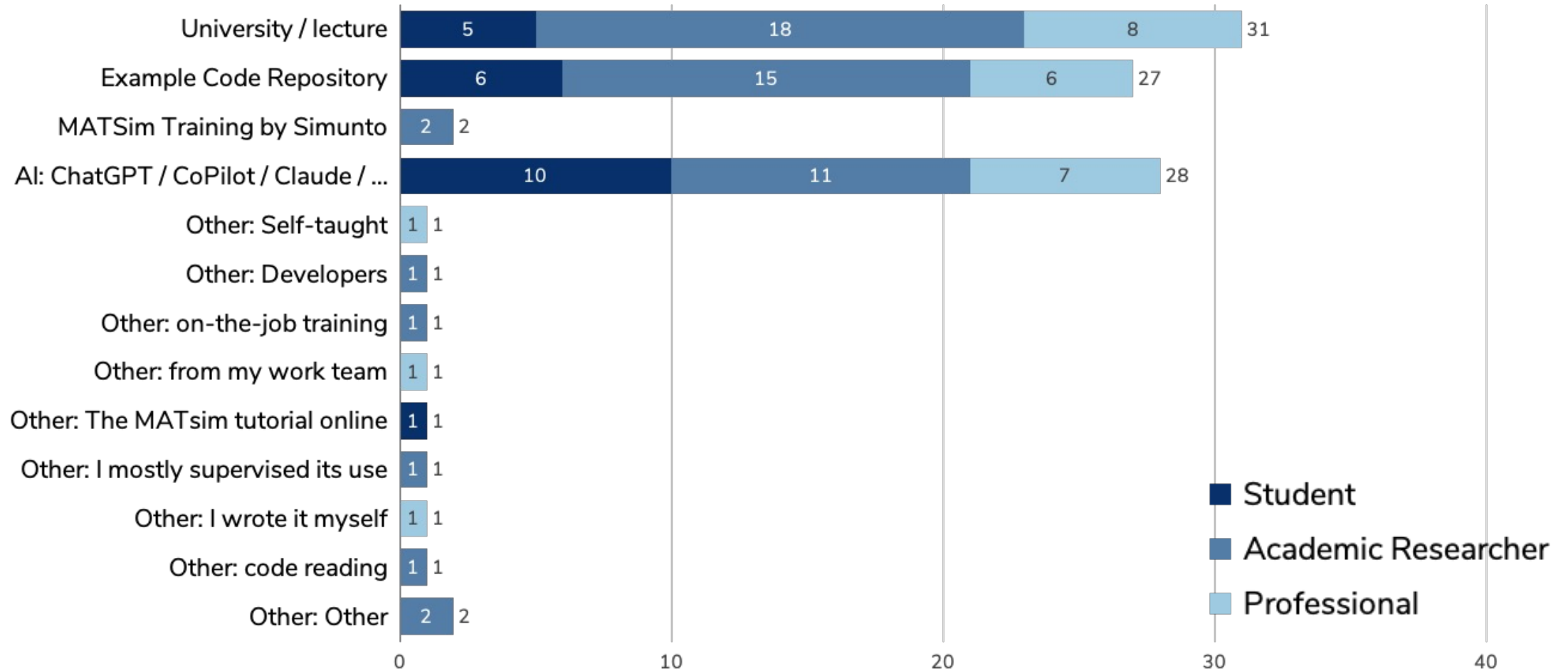


What other tools do you regularly use when working with MATSim or preparing data for it?

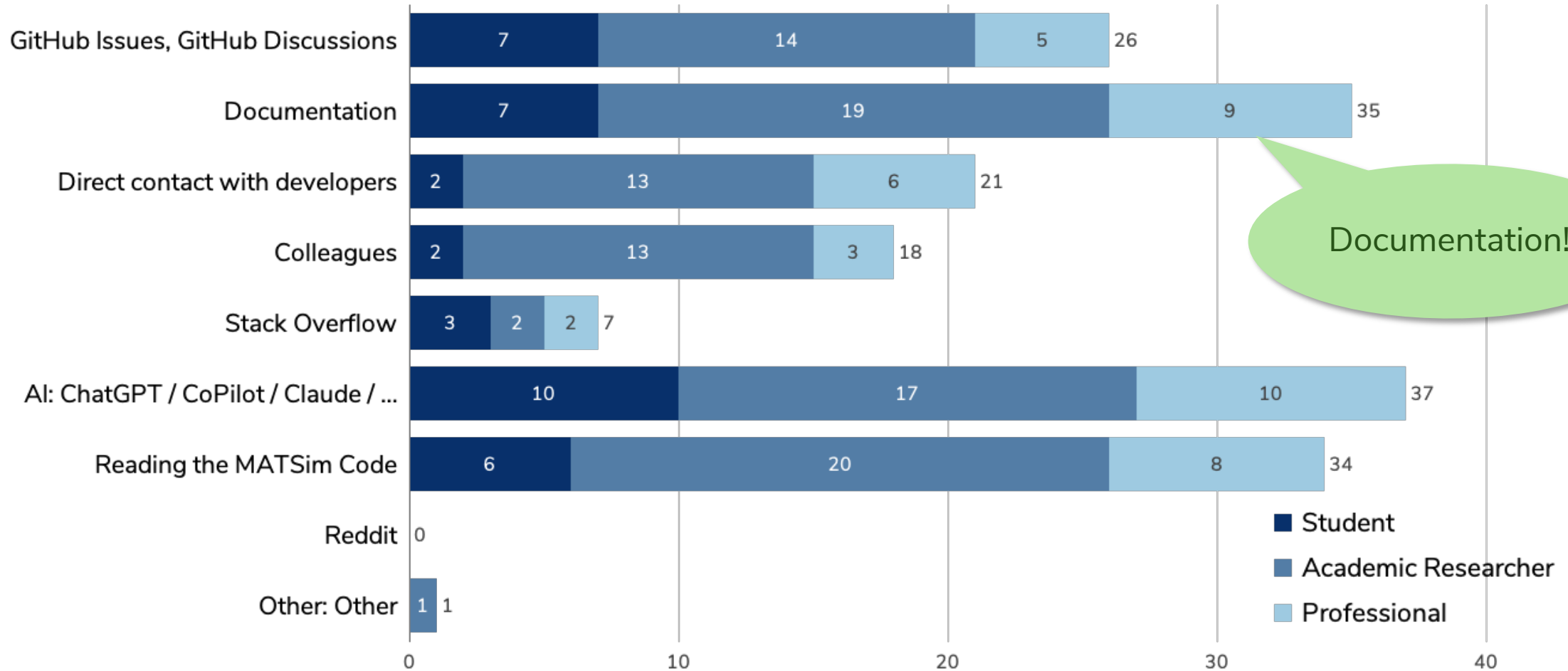


**Development,
Support & Community**

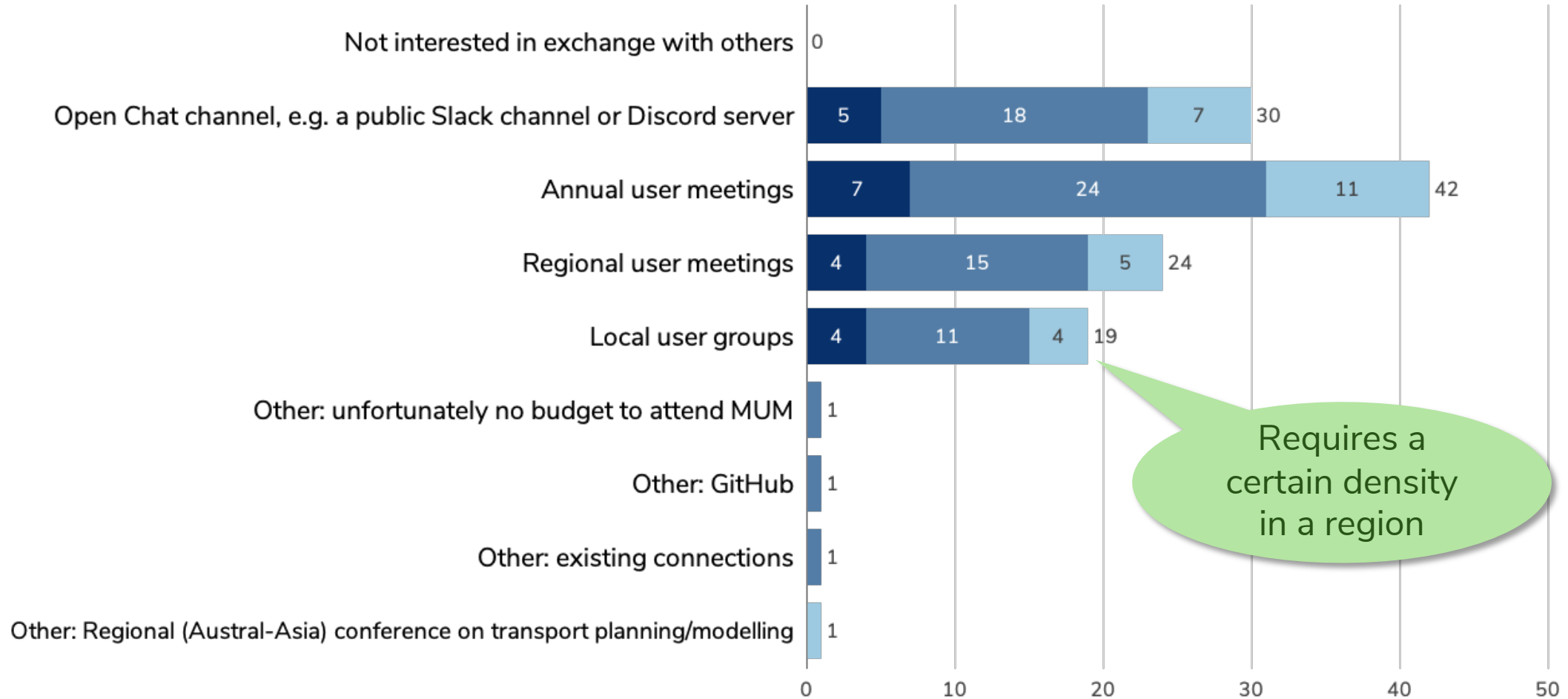
How/Where did you learn how to use/code with MATSim?



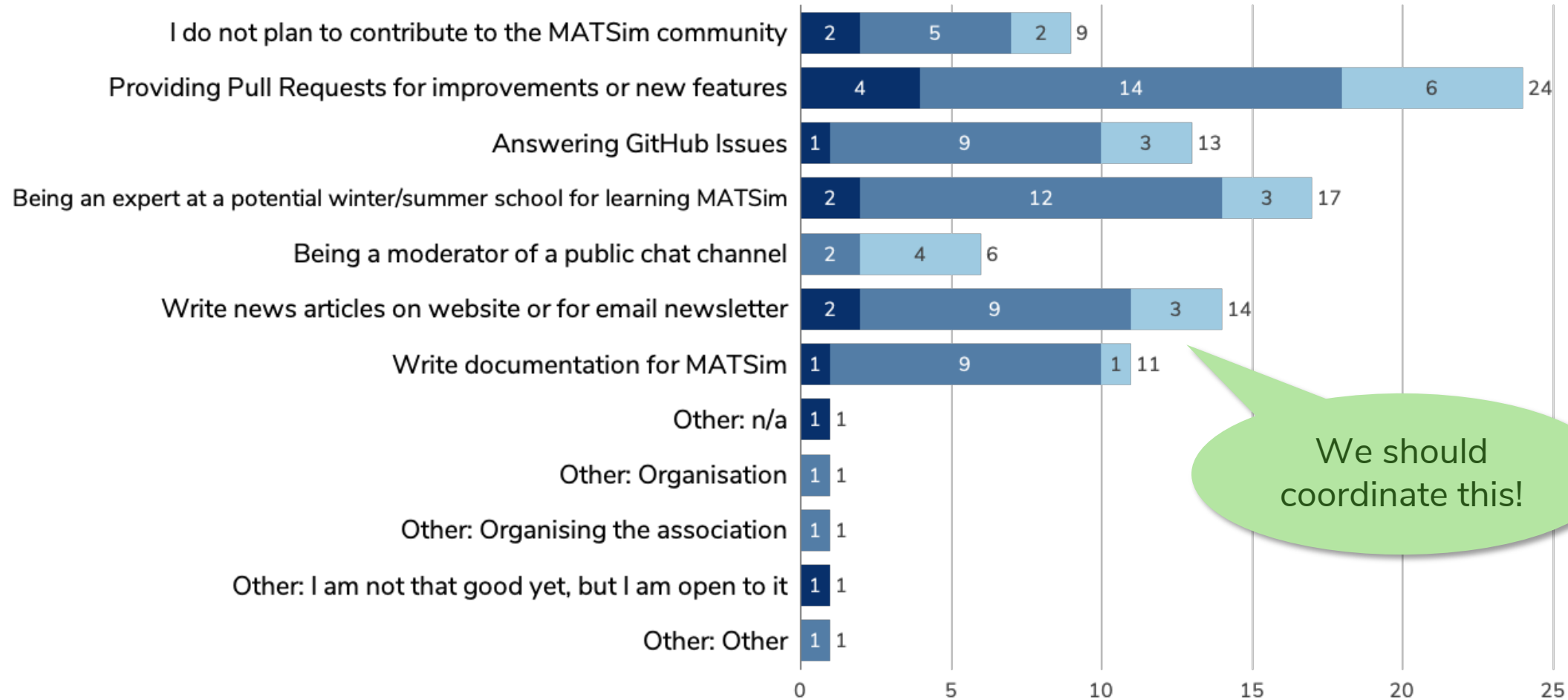
How do you primarily get help with MATSim?



(How) Would you like to get in touch with other MATSim users and developers?



How would you contribute to the MATSim community?

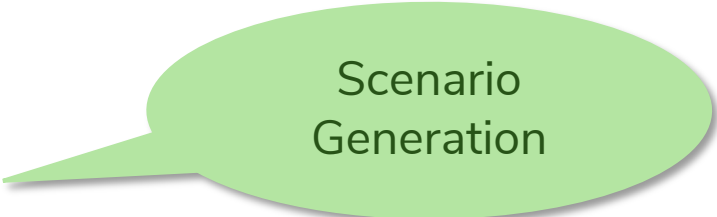


We should coordinate this!

**Development,
Support & Community**

What are your main challenges when using MATSim? (1/5)

- Hard to generate populations & scenario, even harder to calibrate
 - Calibration
 - Getting data
 - Calibrating the model
 - Demand generation as initial input.
-
- The lack of easily starting your own project
 - Learning curve for the new students and team members, keeping the model up to date



Scenario
Generation

What are your main challenges when using MATSim? (2/5)

- Performance, overly complex architecture of the DRT extension making the implementation of new extensions very tedious
- Computation time.
- long running time and high requirements for hardware
- How long it takes to run a sim and how difficult it is to debug something wrong.
- RAM requirements for large scenarios



Performance

What are your main challenges when using MATSim? (3/5)

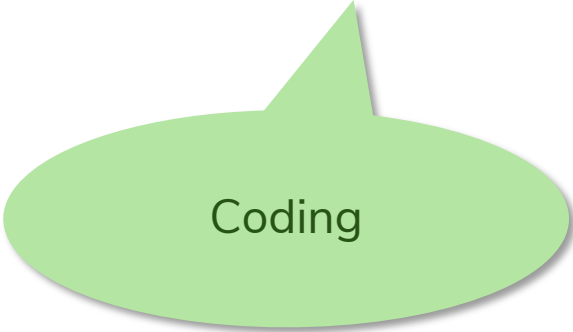
- Finding config options. Finding How-Tos. Learning curve to get new users started. XML!
- XML config files get massive and hard to handle.
- Scaling down to make this easier to do is also very difficult to achieve (a lot of parameters are dependent on the scale).



Configuration

What are your main challenges when using MATSim? (4/5)

- Understanding the complexity/connections between modules.
- A lot of legacy code and little documentation. Understanding features and setting them up often requires a deep dive into the code. Important warnings are drowned in massive log.
- A lot of variables or classes are private and we cannot easily access or replace them without also replacing several others.



Coding

What are your main challenges when using MATSim? (5/5)

- It is complicated as it is a complex modelling software. But that's its strength at the same time.
- Funding
- non-explainability of results, non-stability of results

How would these challenges be best solved? (1/2)

- Solid AI tooling can surely help with improving documentation within the codebase and creating (better) guides and tutorials for modules.
- A central well maintained online destination for documentation / recipes / customization examples / how-tos.
- Documentation/learning videos
- These challenges can be best addressed through better community support, including more active forums, clearer documentation, and accessible guidance for beginners to resolve issues more efficiently.
- I would prefer a MATSim tutorial for beginners. How to set up the entire environment and what it takes to get all files prepared for MATSim.
- Making an easier starting/testing approach
- Provide ready scenarios to people for their cities/universities. This way, they can start working right away.

How would these challenges be best solved? (2/2)

- The MATSim Association should focus less on attracting more users and instead develop tools and procedures for quality assessment.
- Making it less time consuming to run a big simulation would be the number 1, but hard to achieve.
- Possibly PBF, with data handling instructions for major programming languages (related to huge outputs)

Which features would you like to see added to MATSim? (1/2)

- Intersection model
- New traffic dynamics model solving queue model issues
- Parking and time-use models
- Park and ride support.
- Trip purpose-based control on agents' utilities and parameters
- PT stops access and egress time from walking directly after an activity ends, or to activity before its start
- simulate access/egress time

Which features would you like to see added to MATSim? (2/2)

- More rigor, to compete with mainstream modeling packages
- Alternative forms of configuration
- A simple GUI which helps setting up the scenario
- Would love a compact events db that simplifies downstream processing.
- Easier visualization generation
- Guidelines for quality assessment and assurance
- More tools to validate models

Impact & Results

How do you typically use MATSim results in your decision-making processes?

- Customize analytics to customer requirements
- Policy simulations
- Policy assessment and funding.
- passing them over to the stakeholder
- Analysis in form of reports delivered to stakeholders (policy and decisions makers)
- Create outputs presentable to our industry partners to inform decision making
- We build decision support tools incorporating MATSim for emergency management agencies
- Simulate multiple future scenarios and compare the results.
- Input for informative visualizations
- Seeing potential future issues of road and public transport networks that may have not been taken into account just by thinking.

Additional Comments

- Please, let's agree on a code formatting style.
- Would be great if possible to have MUM 27 in Australia! Perhaps with IATBR Conference in Sydney?
- MATSim, and equally its community, is an extremely valuable asset that we should all work to expand.
- I love MATSim! Thanks to the contributors
- I will write a paper using the MATSim results, but currently I am late for September MATSim event :(

**Thanks to all who participated
and to all in the MATSim community!**

See you next year at MUM27!